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## SENTIMENT ANALYSIS SERVICE

### Bao cao phan tich ky thuat doanh nghiep

Pham vi: reverse engineering repository, review kien truc, system audit, va technical due diligence

|                   |                                                   |
|-------------------|---------------------------------------------------|
| Ngày danh gia     | 2026-06-06                                        |
| Thu tu bang chung | Ma nguon, cau hinh, ha tang, test, tai lieu cu    |
| Runtime chinh     | FastAPI + model runtime trong cung process        |
| UI chinh          | Angular SPA sau Nginx                             |
| Chuo ML chinh     | DVC, HuggingFace, PEFT/LoRA, MLflow, ONNX Runtime |

|                   |                    |
|-------------------|--------------------|
| Tech Lead         | Le Quang Tuyen     |
| ML Engineer       | Duong Binh An      |
| AI Engineer       | Do Quoc Trung      |
| Solution Engineer | Duong Hong Quan    |
| Backend Engineer  | Pham Duc Long      |
| UI/UX Engineer    | Nguyen Le Hong Nhi |

#### Phuong phap danh gia

Bao cao duoc tao tu viec doc ma nguon va cau hinh trong repository. Nhung gi khong co trong code duoc danh dau ro la Implemented, Partially Implemented, Mocked, Deprecated, hoac Not Implemented. Moi ket luan deu gioi han theo bang chung ton tai trong repo ngay 2026-06-06.

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## Executive Report

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Repository nay thuc thi mot he thong sentiment analysis co hai mat ro rang: online inference plane va offline ML lifecycle plane. Mat online gom Angular frontend, Nginx reverse proxy, FastAPI backend, va mot model orchestration layer chay trong cung process de phuc vu predict, aspect extraction, sarcasm detection, SHAP explainability, health, metrics, va batch CSV dong bo. Mat offline gom DVC pipeline cho download du lieu, tien xu ly, validation, finetuning, evaluation, ONNX export, benchmark, va MLflow tracking.

Diem manh lon nhat la tinh end-to-end co that o cap repository, abstraction hop ly thong qua `contracts/model_interface.py`, monitoring co mat, va duong di tu training den serving bang artifact ONNX. Diem yeu lon nhat la control plane: khong co authentication, khong co authorization, batch status la mock, evaluation status nam trong memory, va cac path nang van cung chia se mot process.

He thong phu hop nhat voi nhan dien **Hybrid Architecture: modular monolith runtime ket hop workflow ML platform**. Day khong phai he thong microservice phan tan. Backend la mot deployable FastAPI service so huu routing, validation, va orchestration model. Xung quanh no la frontend container, monitoring services, va bo script/pipeline ML cung song trong mot repository.

### Tom tat maturity

Phan loai: Hybrid Architecture / Modular Monolith co workflow ML platform

Muc trung thanh hien tai: Advanced MVP

Production readiness: chi phu hop demo noi bo hoac pilot bi gioi han chat

Ket luan ky thuat tong the: kien truc day du cho hoc thuat va demo, nhung chua du hardening doanh nghiep

## 1 Section 1 - Executive Technical Summary

---

### 1.1 System Purpose

He thong bien van ban tu do thanh ket qua sentiment co cau truc cho hai kieu su dung: tuong tac theo tung cau va xu ly CSV hang loat. Dong thoi repo cung chung minh mot day chuyen AI ung dung hoan chinh: data acquisition, preprocessing, validation, finetuning, evaluation, export artifact, serving, va monitoring.

### 1.2 Technical Scope

- giao dien Angular va luong batch upload trong `app/sentiment-analysis-chatbot`
- backend API trong `src/main.py`
- inference orchestration trong `src/model/`
- data, training, evaluation, export trong `src/data/`, `src/training/`, `src/scripts/`

- Docker, Compose, Prometheus, Grafana, MLflow cho deployment va observability

### 1.3 Technology Overview

| Tang             | Cong nghe                                       |
|------------------|-------------------------------------------------|
| Frontend         | Angular 17, TypeScript, signals, RxJS, Nginx    |
| Backend          | Python 3.11, FastAPI, Uvicorn, Pydantic         |
| Inference        | Transformers, PEFT, ONNX Runtime, SHAP, PyTorch |
| Data va training | Pandas, datasets, scikit-learn, DVC, MLflow     |
| Observability    | Prometheus, Grafana, custom middleware metrics  |
| Dong goi         | Docker multi-stage, Docker Compose              |

### 1.4 Core Capabilities

| Nang luc                        | Trang thai            | Bang chung                                                       |
|---------------------------------|-----------------------|------------------------------------------------------------------|
| Du doan tung cau                | Implemented           | POST /predict trong src/main.py                                  |
| Explainability                  | Implemented           | POST /explain<br>get_shap_explanation() goi                      |
| Batch CSV                       | Implemented           | POST /batch_predict tra ket qua truc tiep                        |
| Batch job tracking ben vung     | Mocked                | GET /batch_status/{job_id} tra payload co dinh                   |
| Trigger evaluation              | Partially Implemented | /evaluate chay background thread va luu state trong memory       |
| Training pipeline               | Implemented           | run_finetuning.py, DVC stages, task configs                      |
| ONNX export                     | Implemented           | export_onnx.py, onnx_exporter.py                                 |
| Monitoring                      | Partially Implemented | Prometheus va Grafana co san; tracing va structured logs chua co |
| Authentication va authorization | Not Implemented       | khong co middleware hay gateway rule nao                         |

### 1.5 Technical Maturity Assessment

Repo da vuot Prototype va MVP co ban vi co test, contracts, monitoring, va deployment manifests. Repo chua dat muc production enterprise vi thieu nhung control cot loi quanh trust boundary, durability, va workload isolation.

### 1.6 System Classification

**Phan loai: Hybrid Architecture.**

Ly do:

- backend la modular monolith: mot FastAPI process so huu routing, validation, model orchestration, va admin-like evaluation trigger

- frontend duoc container hoa rieng va reverse-proxy qua Nginx
- monitoring dung service ben ngoai, nhung runtime ung dung khong tach thanh business services doc lap
- cung mot repo chua ca workflow ML platform voi DVC, MLflow, training, evaluation, va export

## 2 Section 2 - System Architecture Analysis

### 2.1 Architecture Overview

```

1 flowchart LR
2   Browser --> FE[Angular Frontend]
3   FE --> NX[Nginx Reverse Proxy]
4   NX --> API[FastAPI Runtime]
5   API --> LD[Language Detector]
6   API --> INF[BaselineModelInference]
7   INF --> ONNX[ONNX Runtime Session]
8   INF --> HF[HF / PEFT Model Path]
9   INF --> ABSA[Zero-shot ABSA Pipeline]
10  INF --> SHAP[SHAP Explainer]
11  API --> METRICS[/metrics]
12  METRICS --> PROM[Prometheus]
13  PROM --> GRAF[Grafana]
14  API --> EVAL[/evaluate]
15  EVAL --> MLFLOW[MLflow]

```

### 2.2 Layer Assessment

| Layer     | Muc dich                                               | Dependencies                                       | Input                      | Output                               | Failure modes                                   | Rui ro scale                                        |
|-----------|--------------------------------------------------------|----------------------------------------------------|----------------------------|--------------------------------------|-------------------------------------------------|-----------------------------------------------------|
| Frontend  | tuong tac nguoi dung, local state, explain va batch UX | Angular, Client, Storage, browser speech API       | Http- local- browser       | text, CSV, health checks             | HTTP requests, rendered results                 | backend state nam tren browser va payload batch lon |
| Backend   | routing, validation, lifecycle, status surfaces        | FastAPI, Pydantic, global model instance           | HTTP requests, file upload | JSON, re- plain-text errors, metrics | startup load fail, CSV loi, admin route public  | mot process so huu model va route nang              |
| Inference | bien text thanh labels, aspects, sarcasm, SHAP         | Transformers, ONNX Runtime, tokenizers, PEFT, SHAP | text va language           | du doan va giai thich                | artifact thieu, unsupported language, ABSA fail | ABSA va SHAP ton va co cache                        |

| Layer      | Mục đích                                         | Dependencies                            | Input                           | Output                             | Failure modes                              | Rủi ro                                       |
|------------|--------------------------------------------------|-----------------------------------------|---------------------------------|------------------------------------|--------------------------------------------|----------------------------------------------|
| Training   | tao adapters va ket qua evaluation               | datasets, Trainer, PEFT, MLflow, pandas | raw CSVs va params              | adapters, metrics, logs            | dataset drift, imbalance, export chain loi | chi co local hoac single-container execution |
| Data       | download, normalize, validate, prepare eval sets | pandas, XML parser, datasets, DVC       | XML files va HF datasets        | processed CSVs, reports, eval sets | source files thieu, schema loi             | pipeline file-based va governance thu cong   |
| Monitoring | thu thap counters va latency histograms          | prometheus_client, Prometheus, Grafana  | metrics endpoint                | dashboards va alerts               | scrape config sai, dashboard drift         | chi nhin thay cap API, khong co tracing      |
| Deployment | build va ket noi containers                      | Docker, Compose, Nginx, DVC pull        | code, credentials, artifacts    | runnable stack                     | build secret leak, artifact mismatch       | chi co Compose, khong co rollout strategy    |
| Security   | bao ve API, secrets, artifacts, containers       | env vars va default network             | credentials va public endpoints | enforcement policy                 | auth absent, default secret, CORS rong     | trust boundary yeu                           |

### 3 Section 3 - Repository Structure Analysis

#### 3.1 Repository Structure Diagram

```

1 flowchart TD
2   ROOT[Repository Root]
3   ROOT --> APP[app/]
4   ROOT --> CONTRACTS[contracts/]
5   ROOT --> DATA[data/]
6   ROOT --> DOCS[docs/]
7   ROOT --> INFRA[infra/]
8   ROOT --> MODELS[models/]
9   ROOT --> SRC[src/]
10  ROOT --> TESTS[tests/]
11  ROOT --> DVC[dvc.yaml and params.yaml]
12  ROOT --> DOCKER[Dockerfile and docker-compose.yml]

```

#### 3.2 Major Directory Analysis

| Thu muc         | Muc dich                                          | Criticality | Runtime rel-<br>evance             | Dependencies                      | Modification<br>risk | Complexity |
|-----------------|---------------------------------------------------|-------------|------------------------------------|-----------------------------------|----------------------|------------|
| app/            | Angular UI va frontend Docker build               | High        | Direct                             | backend contracts va Ng-inx proxy | High                 | Medium     |
| contracts/      | shared schemas, model interface, errors, examples | High        | Direct                             | FastAPI va tests                  | High                 | Low        |
| data/           | raw, processed, eval, report artifacts            | High        | Indirect run-time; direct training | DVC va scripts                    | High                 | Medium     |
| docs/           | reports va legacy docs                            | Low         | Indirect                           | none at run-time                  | Low                  | Low        |
| infra/          | Prometheus va Grafana provisioning                | Medium      | Direct                             | metrics end-point va Compose DNS  | Medium               | Low        |
| models/         | adapters va ONNX artifacts                        | Critical    | Direct                             | training, export, runtime load    | Critical             | Medium     |
| src/data/       | download, pre-process, validate pipeline          | High        | Training-critical                  | params va raw data                | High                 | Medium     |
| src/model/      | inference, export, evaluation, device logic       | Critical    | Runtime-critical                   | model artifacts va ML libs        | Critical             | High       |
| src/monitoring/ | metric objects va middleware                      | Medium      | Runtime-critical                   | FastAPI middleware chain          | Medium               | Low        |
| src/scripts/    | CLIs cho training, eval, export, benchmark        | High        | Lifecycle-critical                 | training va model modules         | High                 | Medium     |
| src/training/   | reusable training internals va metrics            | High        | Lifecycle-critical                 | Trainer, pandas, sklearn          | High                 | Medium     |
| tests/          | unit va integration evidence                      | Medium      | Indirect                           | pytest va mocks                   | Medium               | Medium     |

### 3.3 Dependency Graph

```

1 flowchart LR
2   APP --> CONTRACTS
3   APP --> MAIN[src/main.py]
4   MAIN --> CONTRACTS
5   MAIN --> MODEL[src/model]
6   MAIN --> MON[src/monitoring]
7   SCRIPTS[src/scripts] --> TRAIN[src/training]
8   SCRIPTS --> DATA[src/data]
9   SCRIPTS --> MODEL

```



```

10  MODEL --> MODELS
11  INFRA --> MAIN

```

### 3.4 Module Interaction Diagram

```

1  flowchart LR
2    ChatUI --> SentimentService
3    SentimentService --> PredictAPI
4    SentimentService --> ExplainAPI
5    SentimentService --> BatchAPI
6    PredictAPI --> BaselineModelInference
7    ExplainAPI --> BaselineModelInference
8    BatchAPI --> BaselineModelInference
9    BaselineModelInference --> OnnxSession
10   BaselineModelInference --> ABSAPipeline
11   BaselineModelInference --> ShapExplainer

```

## 4 Section 4 - Execution Flow Analysis

### 4.1 Workflow Matrix

| Workflow         | Entry point            | Execution path                                                 | Key files                           | Outputs                    | Status      |
|------------------|------------------------|----------------------------------------------------------------|-------------------------------------|----------------------------|-------------|
| Startup          | FastAPI lifespan       | tao ModelConfig, khoi tao BaselineModelInference, preload ABSA | src/main.py, src/model/baselines.py | loaded model, startup fail | Implemented |
| Health check     | GET /health            | dependency injection roi tra payload                           | src/main.py, contracts/schemas.py   | health JSON                | Implemented |
| Prediction       | POST /predict          | validate, detect language, predict single, record latency      | them language_detector.py           | PredictResponse            | Implemented |
| Explanation      | POST /explain          | validate, detect language, build SHAP on demand                | baseline.py                         | ExplainResponse            | Implemented |
| Batch processing | POST /batch_prediction | doc CSV, cap 500 long, thread off sync batch inference         | src/main.py                         | danh sach ket qua          | Implemented |
| Batch status     | GET /batch_status/{id} | tra du lieu co dinh                                            | src/main.py                         | payload hoan thanh gia     | Mocked      |

| Workflow                | Entry point                                     | Execution path                                                           | Key files                                         | Outputs                         | Status                        |
|-------------------------|-------------------------------------------------|--------------------------------------------------------------------------|---------------------------------------------------|---------------------------------|-------------------------------|
| Evaluation              | POST<br>/evaluate<br>va GET<br>/evaluate/status | background task<br>goi logic<br>evaluate.py,<br>status trong<br>memory   | src/main.py,<br>src/model/evaluate.py             | MLflow logs va<br>status fields | Partially<br>Imple-<br>mented |
| Training                | python -m<br>src.scripts.run_finetuning.py      | load raw data,<br>load finetuning<br>oversample, train<br>LoRA adapter   | src/scripts/run_finetuning.py,<br>src/training/   | MLflow run                      | Implemented                   |
| Finetuned<br>evaluation | python -m<br>src.scripts.evaluate_finetuned.py  | load finetuned con-<br>fig, batch finetuned<br>JSON reports              | evaluate_finetuned.py,<br>src/training/metrics.py | metrics JSON                    | Implemented                   |
| ONNX ex-<br>port        | python -m<br>src.scripts.export_onnx.py         | merge PEFT<br>export onnx<br>FP32 + INT8                                 | export_onnx.py,<br>onnx_exporter.py               | ONNX dirs                       | Implemented                   |
| Deployment              | docker<br>compose up                            | build backend,<br>frontend, moni-<br>toring, MLflow,<br>optional trainer | Dockerfiles va Com-<br>pose                       | multi-container<br>stack        | Implemented                   |

## 4.2 Primary Request Sequence

```

1 sequenceDiagram
2   participant U as User
3   participant FE as Angular
4   participant NX as Nginx
5   participant API as FastAPI
6   participant LD as LanguageDetector
7   participant INF as BaselineModelInference
8   U->>FE: gui text
9   FE->>NX: POST /api/predict
10  NX->>API: forward request
11  API->>LD: detect neu thieu lang
12  API->>INF: predict_single(text, lang)
13  INF-->>API: sentiment, confidence, aspects, sarcasm
14  API-->>NX: PredictResponse
15  NX-->>FE: JSON
16  FE-->>U: hien thi ket qua

```

## 4.3 Explain Sequence

```

1 sequenceDiagram
2   participant FE as Angular
3   participant API as FastAPI
4   participant INF as BaselineModelInference
5   participant SHAP as SHAP Explainer

```

```
6 FE->>API: POST /api/explain
7 API->>INF: get_shap_explanation(text, lang)
8 INF->>SHAP: tao explainer va tinh values
9 SHAP-->>INF: tokens, values, base value
10 INF-->>API: SHAPResult
11 API-->>FE: ExplainResponse
```

## 4.4 Training to Serving Sequence

```
1 sequenceDiagram
2   participant DVC
3   participant DATA as Data Pipeline
4   participant TRAIN as Finetuning
5   participant EVAL as Evaluation
6   participant EXP as ONNX Export
7   participant RT as Runtime Image
8   DVC->>DATA: download, preprocess, validate
9   DATA->>TRAIN: normalized CSV inputs
10  TRAIN->>EVAL: adapters and metrics
11  TRAIN->>EXP: trained adapter directories
12  EXP->>RT: ONNX artifacts
13  RT->>RT: startup load
```

## 5 Section 5 - Frontend Analysis

### 5.1 Architecture

Frontend là một Angular application dùng standalone components. State được lưu chủ yếu bằng Angular signals trong `SentimentAnalysisService`. Không có external state manager như `NgRx`.

## 5.2 Components and Services

| Artifact                                   | Vai tro                                                         | Nhan xet                                                   |
|--------------------------------------------|-----------------------------------------------------------------|------------------------------------------------------------|
| <code>app.component.ts</code>              | shell chinh, health polling, render chat history, modal toggles | component lon, giao UI, chua nhieu or-ches-tra-tion        |
| <code>sentiment-analysis.service.ts</code> | API client va message state, explain trigger logic              | service trung tam, cou-pling cao voi pay-load              |
| <code>chat-input.component.ts</code>       | text entry, speech-to-text, language picker, batch button       | nhieu tinh nang; lan-guage sup-port vuot back-end sup-port |
| <code>message-bubble.component.ts</code>   | render ket qua sentiment va SHAP bars                           | explainability UI duoc lam truc tiep                       |
| <code>batch-upload.component.ts</code>     | upload CSV, tom tat ket qua, export CSV                         | batch UX dong bo, khong dung job polling                   |

### 5.3 Frontend Data Flow

```

1 flowchart LR
2   Input[Chat Input]
3   Service[SentimentAnalysisService]
4   API[REST endpoints]
5   State[Signal message state]
6   View[Message bubbles and batch modal]
7   Input --> Service --> API --> State --> View

```

### 5.4 Key Findings

- health polling có that và điều khiển loading overlay
- batch processing UI được implement và kỹ vòng kết qua sync
- explainability UI được implement và vẽ SHAP token contributions
- component phát ra `lang`, nhưng `handleSendText()` và `predict()` gần như bỏ qua giá trị này cho predict thông thường
- language picker hiện nhiều ngôn ngữ, trong khi backend chỉ support `en` và `vi`

### 5.5 Strengths, Weaknesses, Risks

| Loại                     | Danh giá                                                                                                               |
|--------------------------|------------------------------------------------------------------------------------------------------------------------|
| Strengths                | trải nghiệm giao diện chau chuốt, explain UI có that, local persistence, startup có health awareness                   |
| Weaknesses               | top-level component lộn, service gom cả transport và display logic, frontend tests mong, mismatch về language contract |
| Risks                    | UX đã ngôn ngữ dễ gây hiểu nhầm, coupling cao với bot message formatting, state chỉ nằm trên browser                   |
| Maintainability concerns | inline template lộn, UI logic và transport logic trộn lẫn, mapping payload nằm trong code tạo chuỗi hiển thị           |

## 6 Section 6 - Backend Analysis

### 6.1 FastAPI Architecture

Backend là một FastAPI application đơn lẻ trong `src/main.py`. Nó sử dụng:

- lifespan hook để khởi tạo model
- dependency injection cho model instance
- Pydantic schemas cho request và response
- middleware để ghi Prometheus metrics

- background tasks chi cho evaluation

## 6.2 API Surface

| Method | Path                   | Muc dich                          | Validation                                               | Failure modes                         | Performance notes                      |
|--------|------------------------|-----------------------------------|----------------------------------------------------------|---------------------------------------|----------------------------------------|
| GET    | /health                | readiness va supported languages  | model dependency phai resolve                            | model not loaded se 503 tu dependency | re                                     |
| POST   | /predict               | du doan tung cau                  | text 1..2000 qua schema                                  | unsupported language, model error     | ABSA va sarcasm tang latency           |
| POST   | /explain               | SHAP explanation                  | cung schema voi predict                                  | SHAP hoac tokenizer fail              | latency cao                            |
| POST   | /batch_predict         | cham diem CSV dong bo             | CSV hop le, co cot <code>text</code> , gioi han 500 dong | bad CSV, dong rong, inference fail    | thread-offloaded nhung van CPU-bound   |
| GET    | /batch_status/{job_id} | cham diem cho batch job status    | chi co path string                                       | khong co; tra gia tri co dinh         | mocked                                 |
| POST   | /evaluate              | trigger evaluation tu runtime     | chan concurrent runs bang in-memory flag                 | thieu processed data, MLflow fail     | heavy admin-like task trong public app |
| GET    | /evaluate/status       | expose in-memory evaluation state | none                                                     | reset khi restart process             | re nhung khong ben                     |
| GET    | /metrics               | Prometheus scrape endpoint        | none                                                     | metrics library fail                  | re                                     |

## 6.3 Contracts and Validation

Validation manh o JSON edge nho Pydantic schemas. Validation file yeu hon: batch upload chi kiem CSV doc duoc va co cot `text`; khong ep MIME type, file size ngay trong FastAPI, hay content sanitization ngoai viec trim text.

## 6.4 Middleware and Background Tasks

Chi co mot middleware la `monitor_middleware`. Background task chi dung cho evaluation, khong dung cho batch scoring hay explainability.

# 7 Section 7 - AI Inference Analysis

## 7.1 Inference Pipeline

```

1 flowchart LR
2   Text --> Lang[Language Resolution]
3   Lang --> Sent[Sentiment Model]
```

```

4 Sent --> Sarc[Sarcasm Model or Fallback False]
5 Sent --> ABSA[Zero-shot ABSA]
6 Sent --> Resp[PredictionResult]
7 Text --> SHAP[SHAP Explainer on Demand]

```

## 7.2 Stage-by-Stage Assessment

| Stage               | Input                    | Transformation                                      | Output                                                      | Cost impact           | Failure modes                                                     |
|---------------------|--------------------------|-----------------------------------------------------|-------------------------------------------------------------|-----------------------|-------------------------------------------------------------------|
| Model loading       | config mode<br>va paths  | chon ONNX, ONNX INT8, finetuned, hoac baseline path | live model/session state                                    | startup nang          | artifact thieu, export khong tuong thich                          |
| Tokenizer           | text list                | tokenization, truncation, padding                   | tensors hoac NumPy arrays                                   | trung binh            | tokenizer thieu hoac mismatch                                     |
| Language detection  | optional lang<br>va text | heuristic nhan dien dau tieng Viet                  | <del>LanguageDetection</del> <b>LanguageDetectionResult</b> | <del>chi phi</del>    | detect sai, dac biet voi text ngan                                |
| Sentiment inference | tokenized text           | PyTorch forward pass hoac ONNX session run          | class probabilities                                         | chi chinh             | provider mismatch, model error                                    |
| Sarcasm detection   | text                     | them mot lan predict hoac default false             | boolean flag                                                | tang chi phi tung mau | sarcasm model thieu thi silently degrade ve false trong ONNX mode |
| ABSA                | text                     | zero-shot detect aspect va sentiment-per-aspect     | aspect list                                                 | rat cao               | pipeline fail thi tra aspects rong                                |
| SHAP                | text                     | tao explainer va perturbation sampling              | token contributions                                         | cao nhat              | khong co tokenizer, latency, memory pressure                      |

## 7.3 Important Inference Findings

- ModelConfig support baseline, finetuned, onnx, va onnx\_int8
- runtime mac dinh la onnx thong qua fallback cua MODEL\_MODE
- ABSA khong phai mot model aspect duoc train rieng; no la zero-shot classification tren tap labels co cau hinh
- language detection cho tieng Viet chi la heuristic
- trong batch mode, ABSA bi bo qua de giu throughput

## 8 Section 8 - ML Pipeline Analysis

### 8.1 DVC Pipeline Overview

```

1 flowchart LR
2   download --> preprocess --> validate --> evaluate_baseline
3   download_sarcasm --> finetune_sarcasm --> export_onnx_sarcasm
4   download_sentiment --> prepare_eval --> finetune_sentiment --> evaluate_finetuned -->
   export_onnx_sentiment --> benchmark_onnx

```

### 8.2 Stage Assessment

| Stage              | Mục đích                                 | Inputs                                       | Outputs                          | Artifacts                        | Failure modes                        |
|--------------------|------------------------------------------|----------------------------------------------|----------------------------------|----------------------------------|--------------------------------------|
| download           | parse XML thành raw CSVs<br>SemEval XML  | XML local trong<br>data/external/semeval2014 | raw sentence va as-<br>pect CSVs | data/raw/*.csv                   | XML thiếu hoặc schema drift          |
| preprocess         | normalize va split dataset               | raw sentence va aspect CSVs                  | processed CSVs                   | data/processed/                  | bad columns, transform fail          |
| validate           | quality gate processed data              | processed CSVs va params                     | JSON quality report              | data/reports/quality_report.json | quality report fail                  |
| download_sarcasm   | tweet-eval irony data                    | HuggingFace dataset                          | sarcasm raw CSV                  | data/raw/sarcasm/                | network hoac dataset change          |
| download_sentiment | lay EN va VI sentiment data              | HuggingFace parquet sources                  | hai raw CSV                      | data/raw/sentiment/              | dataset, source fail                 |
| prepare_eval       | tao held-out eval CSVs                   | raw sentiment CSVs                           | mixed-lang va VI probe sets      | data/eval/                       | raw data missing                     |
| finetune           | train adapters LoRA                      | raw training CSVs va params                  | adapter dirs va MLflow runs      | models/adapters/                 | imbalance, training fail             |
| evaluate_finetuned | chinh fairness-aware reports             | adapters va raw/eval data                    | JSON metrics                     | reports/*.json                   | metric mismatch hoặc missing adapter |
| export_onnx        | doi adapter thành serving artifact       | finetuned adapter path                       | FP32 va INT8 ONNX dirs           | models/onnx/                     | merge/export fail                    |
| benchmark_onnx     | so sanh cac mode ONNX va finetuned model | generated ONNX va finetuned model            | benchmark JSON                   | reports/onnx_benchmark.json      | benchmark failure                    |

## 9 Section 9 - Data Flow Analysis

### 9.1 Data Lineage

```

1 flowchart LR
2   Raw[Raw Data]
3   Prep[Processing]
4   Val[Validation]
5   Train[Training]
6   Eval[Evaluation]

```



```
7   Export[ONNX Export]
8   Load[Runtime Load]
9   Predict[Prediction]
10  Raw --> Prep --> Val --> Train --> Eval --> Export --> Load --> Predict
```

## 9.2 Artifact Lifecycle

| Artifact type                 | Vi tri                                                                    | Vong doi                                                                       |
|-------------------------------|---------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| SemEval raw CSVs              | <code>data/raw/sentences.csv</code> , <code>data/raw/aspects.csv</code>   | sinh tu XML, duoc preprocessing su dung                                        |
| Processed training data       | <code>data/processed/</code>                                              | sinh boi pipeline, duoc baseline evaluation va runtime-side evaluation su dung |
| Sentiment va sarcasm raw CSVs | <code>data/raw/sentiment_*.csv</code> , <code>data/raw/sarcasm.csv</code> | download de fine-tune va evaluate                                              |
| Eval CSVs                     | <code>data/eval/</code>                                                   | sinh boi <code>prepare_eval.py</code> , duoc fine-tuned evaluation su dung     |
| Adapters                      | <code>models/adapters/</code>                                             | sinh boi finetuning, duoc evaluate va export ONNX                              |
| ONNX models                   | <code>models/onnx/</code>                                                 | su dung sinh boi export, duoc runtime image va inference su dung               |
| MLflow runs                   | <code>mlruns/</code> hoac tracking URI ben ngoai                          | sinh trong 17 training va evaluation.                                          |

## 10 Section 10 - API Analysis

### 10.1 Request and Response Summary

| Endpoint                   | Auth status | Schema source                    | Ghi chu                                     |
|----------------------------|-------------|----------------------------------|---------------------------------------------|
| GET /health                | public      | HealthResponse                   | bao supported languages tu model config     |
| POST /predict              | public      | PredictRequest/PredictResponse   | business endpoint quan trong nhat           |
| POST /explain              | public      | ExplainRequest/ExplainResponse   | dat, khong co throttling                    |
| POST /batch_predict        | public      | multipart + BatchPredictResponse | dong bo du docs o noi khac de nhu async job |
| GET /batch_status/{job_id} | public      | BatchStatusResponse              | mocked endpoint                             |
| POST /evaluate             | public      | ad hoc JSON                      | admin-like action bi expose                 |
| GET /evaluate/status       | public      | ad hoc JSON                      | state chi nam trong memory                  |
| GET /metrics               | public      | Prometheus text                  | scrape endpoint khong auth                  |

## 11 Section 11 - Deployment Analysis

### 11.1 Deployment Diagram

```

1 flowchart TD
2   subgraph Compose
3     FE[frontend :80]
4     API[fastapi_app :8000]
5     PROM[prometheus :9091]
6     GRAF[grafana :3000]
7     MLF[mlflow :5005]
8     TR[trainer profile]
9   end

```

```

10 FE --> API
11 PROM --> API
12 GRAF --> PROM
13 API --> MLF
14 TR --> MLF

```

## 11.2 Container and Runtime Notes

- backend Dockerfile la multi-stage: build dependencies, DVC model pull, final runtime
- runtime image pre-download HuggingFace assets roi bat offline mode
- runtime image dung DVC build args cho DagsHub credentials; chay duoc nhung rui ro
- frontend container proxy /api/ ve backend va mo rong upload/time limits
- Compose co Prometheus, Grafana, MLflow, va mot training profile tuy chon

## 11.3 Deployment Constraints

- Compose la duong orchestration duy nhat duoc commit
- khong co manifests Kubernetes hay managed deployment
- khong co rolling release, canary, hay blue-green workflow
- startup phu thuoc model artifacts va downloaded base assets phai ton tai va tuong thich

# 12 Section 12 - Observability Analysis

## 12.1 Implemented Coverage

| Area                           | Status                | Bang chung                                                          |
|--------------------------------|-----------------------|---------------------------------------------------------------------|
| Request metrics                | Implemented           | counters va histograms trong <code>src/monitoring/metrics.py</code> |
| Model inference latency metric | Implemented           | <code>MODEL_INFERENCE_LATENCY</code> theo language                  |
| Prometheus scrape config       | Implemented           | <code>infra/prometheus/prometheus.yml</code>                        |
| Alert rules                    | Partially Implemented | chi co 2 alerts: high 5xx va high latency                           |
| Grafana provisioning           | Implemented           | datasource va dashboard provisioning files ton tai                  |
| Structured logging             | Partially Implemented | Python logging co, nhung khong co request log format ro rang        |
| Tracing va correlation IDs     | Not Implemented       | khong co OpenTelemetry hay request-id propagation                   |
| Alert routing va paging        | Not Implemented       | khong co Alertmanager config hay notification target                |

## 13 Section 13 - Security Analysis

### 13.1 Security Findings

| Finding                         | Severity | Bang chung                                          | Tac dong                                                   | Khuyen nghi                                          |
|---------------------------------|----------|-----------------------------------------------------|------------------------------------------------------------|------------------------------------------------------|
| Khong co authentication         | Critical | khong co auth dependency hay gateway auth           | public co the lam dung predict, explain, evaluate, metrics | them gateway va app authentication                   |
| Khong co authorization          | Critical | public va admin-like routes nam chung               | heavy hoac privileged actions bi expose                    | tach /evaluate va operator functions                 |
| Permissive CORS                 | High     | allow_origins=["*"]                                 | moi browser origin deu goi duoc API                        | gioi han allowlist theo moi truong                   |
| Grafana password mac dinh       | High     | GF_SECURITY_ADMIN_PASSWORD                          | DBAdmin bi truy cap trai phep                              | dua vao secret va rotate                             |
| Build-time secrets cho DVC pull | High     | Docker build args cho DagsHub credentials           | credentials de lo qua build system hoac cache              | dung runtime secret injection hoac artifact registry |
| Khong co rate limiting          | High     | khong co throttle logic                             | denial-of-wallet tren explain hoac batch                   | them quota va rate controls                          |
| File upload controls yeu        | Medium   | chi check CSV parse va cot text                     | upload qua lon hoac sai dinh dang gay stress backend       | ep size, MIME, va row policy                         |
| Artifact integrity ngam dinh    | Medium   | khong co checksum hay signature validation          | co the serve nham artifact                                 | them checksum va promotion gate                      |
| Container hardening han che     | Medium   | images mac dinh, khong thay non-root runtime config | attack surface rong hon                                    | chay non-root va scan image                          |

## 14 Section 14 - Performance Analysis

### 14.1 Performance Profile

- startup cost bi chi pho boi model/session loading va ABSA preload
- cost du doan tung cau o muc vua trong ONNX mode, nhưng tang manh neu co ABSA va sarcasm
- explain cost la cao nhat vi SHAP duoc tao on demand
- batch throughput duoc cai thien nhờ chunked inference va bo qua ABSA, nhưng van

single-host va in-process

## 14.2 Evidence-backed Observations

- benchmark tests assert muc tieu it nhat 100 samples/sec cho sentiment-only batch throughput
- Prometheus warning threshold xem average inference tren 500 ms la degrade
- Docker Compose tang CPU va memory limits cho backend de phuc vu inference
- exact startup va memory telemetry chua duoc commit thanh runtime measurements

## 15 Section 15 - Failure Mode Analysis

---

### 15.1 Failure Tree

```
1 flowchart TD
2   A[Startup failure] --> A1[missing model artifacts]
3   A --> A2[tokenizer or HF asset mismatch]
4   B[Prediction failure] --> B1[unsupported language]
5   B --> B2[model exception]
6   C[Explain failure] --> C1[SHAP cost or memory pressure]
7   C --> C2[tokenizer unavailable]
8   D[Batch failure] --> D1[invalid CSV]
9   D --> D2[long-running CPU-bound inference]
10  E[Training failure] --> E1[missing raw data]
11  E --> E2[dataset imbalance or bad labels]
12  F[Deployment failure] --> F1[bad credentials or DVC pull]
13  F --> F2[Compose config drift]
```

## 15.2 Failure Summary

| Area             | Trieu chung                         | Recovery path                                                                          |
|------------------|-------------------------------------|----------------------------------------------------------------------------------------|
| Startup          | health 503 hoac container crash     | xac minh artifact paths, mode, tokenizer assets, image build inputs                    |
| Prediction       | 400/500 responses hoac aspects rong | xac minh language handling va model load, xem warning logs                             |
| Explanation      | response rat cham hoac fail         | tach explain path, throttle usage,kiem tra tokenizer va SHAP deps                      |
| Batch processing | request timeout hoac parser error   | xac minh CSV input, giam payload size, can worker architecture                         |
| Evaluation       | status stuck hoac reset sau restart | nhan ra state nam trong memory, rerun evaluation, chuyen sang job tracking ben vung    |
| Training         | stage fail hoac artifact rong       | rerun DVC stages,kiem tra raw files, xac minh params va task configs                   |
| Deployment       | build hoac stack launch fail        | xem Compose logs, secret inputs, DVC pull behavior, network aliases                    |
| Monitoring       | dashboard rong hoac khong co alert  | test <code>/metrics</code> , Prometheus target status, Grafana datasource provisioning |

## 16 Section 16 - Technical Debt Analysis

| Debt class             | Rank     | Bang chung                                                                                              |
|------------------------|----------|---------------------------------------------------------------------------------------------------------|
| Security debt          | Critical | khong co auth/authz, CORS rong, Grafana password mac dinh, evaluate public                              |
| Operational debt       | High     | batch status mock, evaluation state trong memory, khong co alert routing, khong co tracing              |
| Architectural debt     | High     | mot process so huu public API va ML orchestration nang; khong co worker isolation cho explain hay batch |
| Frontend contract debt | High     | UI language selector vuot backend support; explicit lang khong di het predict path                      |
| ML governance debt     | Medium   | artifact integrity va promotion checks moi o muc ngam dinh                                              |
| Testing debt           | Medium   | Python coverage tot, nhung frontend va security tests mong                                              |
| Documentation debt     | Medium   | README va docs cu lech implementation o endpoint va batch behavior                                      |

## 17 Section 17 - Scalability Analysis

### 17.1 Scalability View

- application scalability bi gioi han boi mot in-process model owner va khong co external queue
- model scalability bi gioi han boi chi phi ABSA va SHAP, khong chi sentiment inference

- infrastructure scalability bị giới hạn trong khuôn khổ Compose hơn là managed orchestration
- operational scalability bị giới hạn bởi access control yêu cầu và workflow state không bền
- data scalability tạm đủ cho kiểu CSV và DVC hiện tại, nhưng governance vẫn thiếu
- team scalability bị ảnh hưởng bởi kiến thức tập trung vào `src/model/`, `src/scripts/`, và deployment config

## 18 Section 18 - Architectural Assessment

| Dimension             | Score | Justification                                                             |
|-----------------------|-------|---------------------------------------------------------------------------|
| Architecture quality  | 7/10  | layering rõ, artifact flow hợp lý, nhưng runtime vẫn tập trung quá nhiều  |
| Maintainability       | 6/10  | backend tách module khá ổn; frontend components lộn xộn và mixed concerns |
| Reliability           | 5/10  | có health và tests, nhưng durability boundaries yếu                       |
| Security              | 3/10  | thiếu nhiều control chi tiết sản xuất                                     |
| Performance           | 6/10  | ONNX và batching giúp nhiều; explain và ABSA vẫn chưa                     |
| Observability         | 6/10  | có metrics và dashboards; tracing và alert routing chưa có                |
| Extensibility         | 7/10  | model mode abstraction và task configs dễ mở rộng                         |
| Operational readiness | 5/10  | local integrated stack tốt; enterprise controls chưa đủ                   |
| Testing maturity      | 7/10  | Python coverage tốt; frontend và security tests hạn chế                   |
| ML lifecycle maturity | 8/10  | DVC, MLflow, training, evaluation, export, benchmark đều có               |

## 19 Section 19 - Recommendations

| Horizon    | Van đề                                | Bang chung                                            | Priority | Complexity | Expected outcome                                      |
|------------|---------------------------------------|-------------------------------------------------------|----------|------------|-------------------------------------------------------|
| Immediate  | public API và admin routes không auth | không có auth trong backend hay gateway               | Critical | Medium     | có trust boundary rõ ràng và pilot an toàn hơn        |
| Immediate  | batch status contract gay hieu nham   | mocked <code>/batch_status</code> endpoint            | High     | Low        | loại bỏ async semantics giả hoặc triển khai jobs thật |
| Immediate  | frontend/backend language drift       | UI hiện nhiều language, predict path bỏ explicit lang | High     | Medium     | tăng độ dung và user trust                            |
| Short-term | tách heavy evaluation path            | <code>/evaluate</code> chạy trong public app process  | High     | Medium     | giảm abuse và ổn định runtime                         |
| Short-term | harden secret handling                | build args và Grafana password mac dinh               | High     | Medium     | giảm credential exposure                              |



| Horizon     | Van de                                               | Bang chung                                            | Priority | Complexity | Expected outcome                                     | out- |
|-------------|------------------------------------------------------|-------------------------------------------------------|----------|------------|------------------------------------------------------|------|
| Short-term  | them rate limiting va file controls                  | khong co throttling; upload validation toi thieu      | High     | Medium     | resilience tot hon khi bi misuse                     |      |
| Medium-term | dua batch va explain nang sang background workers    | hien tai la synchronous in-process CPU work           | Medium   | High       | latency de doan duoc va scale tot hon                |      |
| Medium-term | them artifact integrity checks va promotion workflow | khong co checksum hay signed promotion path           | Medium   | Medium     | model release an toan hon                            |      |
| Long-term   | vuot qua Compose-only deployment                     | khong co managed rollout strategy hay secret platform | Medium   | High       | operational scalability va deployment safety tot hon |      |

## 20 Section 20 - Final Technical Conclusion

### 20.1 Major Strengths

- full repository-level ML lifecycle duoc implement that
- runtime va training concerns duoc tach du ro cho doi nho
- contracts, tests, monitoring, va export paths giup system audit duoc
- ONNX serving path, batch optimization, va MLflow integration la nhung quyet dinh thuc dung

### 20.2 Major Weaknesses

- khong co authentication hay authorization
- batch async semantics bi mock, evaluation state khong ben
- frontend/backend contract drift quanh language handling va batch behavior
- heavy inference features chia se cung mot public process, khong co workload isolation

### 20.3 Critical Risks

- su dung trai phep cac endpoints dat tien hoac co tinh chat quan tri
- van hanh hieu sai do batch status khong phan anh thuc te
- performance sup do khi traffic nghieng ve explain hoac batch
- artifact va secret handling yeu trong build va deployment

## 20.4 Production Readiness and Final Verdict

Chi dua tren bang chung implementation, he thong **chua san sang cho production enterprise**. Tuy vay, day la mot Advanced MVP dang tin va la mot nen tang demo hoac pilotkiem soat tot. Kien truc chay duoc end-to-end, ML lifecycle co that, va repository co gia tri ky thuat ro rang. Khoang cach hien tai khong nam o cho thieu feature; no nam o cho thieu control plane can thiet de van hanh cac feature do mot cach an toan, ben vung, va mo rong duoc.